

WEI WANG

Email: wwang93@vols.utk.edu | [GitHub](#) | [LinkedIn](#)
Knoxville, TN, USA

PROFESSIONAL SUMMARY

Ph.D. Candidate in Learning Design and Technology, co-founder of askabout.edu - an evidence-informed decision-support venture for education currently in formation - and Student Principal Investigator researching how generative AI can support critical thinking, AI literacy, and evidence-informed decision making. My work combines design-based research, mixed methods, learning analytics, and AI/ML development to build and evaluate educational systems, including evidence-grounded RAG tools and national interactive data platforms. I co-developed the AIREA Data Explorer and am first author of a forthcoming CRC Press book on computational analysis of educational data. Previously, I conducted mixed-method evaluations of large-scale smart-education initiatives at iFLYTEK's National Cognitive Intelligence Key Laboratory. Technical expertise includes Python, R, LLM systems, NLP, and interactive data visualization.

EDUCATION

University of Tennessee, TN, USA	08/2023 – present
Ph.D. Learning Design and Technology, Education Advisor: Dr. Joshua Rosenberg, Dr. Enilda Romeroh-Hall Johnson University, TN, USA	06/2020 – 12/2021
M.A., Education Technology Anhui Normal University, Wuhu, China	09/2019 – 06/2022
M.S., Education Technology Longyan University, Longyan, China	09/2012 – 06/2016
B.S., Software Engineering	

PUBLICATIONS

- **Wang, W.**, Rosenberg, J. M., & Koehler, M. J. (2025). Trends in AI Curriculum Materials: Insights from Data Mining a Teacher Marketplace. [EdArXiv Preprint](#).
- **Wang, W.**, Pritchard, C., & Rosenberg, J. M. (2025). *Educational Video Transcript Analysis with LLMs: Improving Entity Recognition and Qualitative Insights*. [EdArXiv Preprint](#).
- Ping, H., **Wang, W.**, Xie, Y., Lv, S., Li, J., & Weng, L. (2025, April). Generative AI Dependency in Higher Education: Investigating Continuance Intention, Cognitive Response, and Creativity. In 2025 7th International Conference on Computer Science and Technologies in Education (CSTE) (pp. 615-621). IEEE. <https://doi.org/10.1109/CSTE64638.2025.11092242>
- Allman, B., Kimmons, R., **Wang, W.** et al. Trends and Topics in Educational Technology, 2024 Edition. *TechTrends* (2024). <https://doi.org/10.1007/s11528-024-00950-5>
- Sarah Narvaiz, Qinyun Lin, Joshua M. Rosenberg, Kenneth A. Frank, Spiro J. Maroulis, **Wei Wang**, Ran Xu (2024). Konfound: An R Sensitivity Analysis Package to Quantify the Robustness of Causal Inferences. *Journal of Open Source Software*, 9(95), 5779, <https://doi.org/10.21105/joss.05779>
- Ye, F.-z., & **Wang, W.** (2023). Returning to the main position of classroom education: A study on the teaching efficacy of smart classrooms and its influencing factors. *Journal of Bingtuan Education Institute*, 33(2), 57-65. (CNKI)
- **Wang, W.**, Duan, Y., Wang, Q., & Liu, H. (2021). An ECM-ISC Based on College Students' Continued Learning Intention toward E-Learning Space Post COVID-19. *Open Journal of Social Sciences*, 9, 377-395. DOI: [10.4236/jss.2021.912026](https://doi.org/10.4236/jss.2021.912026)
- **Wang, W.** (2020). Individual-construction strategy of school informatization based on MOOC. *China Education Technology & Equipment*. (CNKI)

Book

Wang, W., Akcaoglu, M., Rosenberg, J. M., & Kellogg, S. (under contract, anticipated

publication late 2026). [*Computational social science cookbook with R: A practical guide*](#) (a volume in the *Chapman & Hall/CRC Big Data Series*). CRC Press.

RESEARCH EXPERIENCE

Project: [askabout.edu](#) - Evidence-Informed Decision Support for Education

Role: Co-founder | Company formation in progress

- Designed and deployed AskAboutEdu, an open-source AI interface that connects large language models (LLMs) with structured educational research evidence, enabling users to query intervention reports and receive answers grounded in primary sources.
- Built an end-to-end retrieval-augmented generation (RAG) pipeline using Python, Streamlit, and sentence-level embeddings to index and retrieve evidence from What Works Clearinghouse (WWC) intervention reports.
- Developed a reproducible data processing workflow to extract, normalize, and chunk long-form PDF research reports into structured JSON corpora, preserving document-level metadata for transparent citation and traceability.
- Implemented evidence-constrained prompting strategies that require the model to answer only from retrieved excerpts, reducing hallucination and enabling explicit source attribution at the chunk level.
- Designed the interface to support researchers, policymakers, and educators in exploring educational evidence interactively, serving as a prototype for transparent, explainable, and evidence-based AI systems in applied education research.

Project: Next Level Teaching Blueprint — Scaling AI-Assisted Critical Thinking Across Research Methods Courses

University of Tennessee | PI: Dr. Fatima T. Zahra | Role: Student Principal Investigator (Student PI) & Graduate Researcher

- Co-led a multi-phase design-based research initiative examining how generative AI can function as a structured discourse partner to strengthen higher-order thinking, critical thinking, and AI literacy in undergraduate and graduate research methods courses.
- Contributed to the design and implementation of constraint-first, dialogic learning activities that require students to articulate their initial reasoning, interrogate assumptions through interaction with an LLM, revise their responses, and document metacognitive reflections.
- Developed an integrated mixed-methods research workflow linking pre/post measures of disciplinary learning, AI literacy, and critical thinking with student learning diaries, AI chat logs, assignment artifacts, classroom observations, and faculty feedback across the Spring 2026 CEHHS Phase 1 pilot.
- Conducted quantitative analyses for the Spring 2025 pre-pilot study, including data validation, demographic reporting, pre/post comparisons, and the production of publication-ready tables and visualizations for statistical learning and AI literacy outcomes.
- Designed computational analyses of student–AI discourse, including interaction-sequence modeling, dialogue-act classification, linguistic markers of learner authority and deference, and learner positioning of AI as an answer generator, validator, or co-thinker.
- Collaborated with an interdisciplinary research team and participating faculty to translate empirical findings into scalable pedagogical principles, faculty implementation resources, and scholarly publications, including a Phase 1 manuscript targeted for CHI 2027.

Project: Elevating Higher Education with AI — National Webinar Series & Open Course (URL: [AI and ME](#))

University of Tennessee, Knoxville (CEHHS), in collaboration with Teaching & Learning Innovation (TLI) | Role: Instructor & Project Contributor PI: Dr. Yukyeong Song

- Supported the design and delivery of a national professional development initiative focused on the responsible and effective use of generative AI in higher education teaching, assessment, and faculty workflows.
- Coordinated implementation of a multi-session (5-session) live webinar series, including session logistics, speaker support, participant communication, and post-session resource management to ensure high-quality, interactive learning experiences.
- Produced, edited, and organized webinar transcripts and learning artifacts to improve accessibility and support reuse of content for professional learning and research-informed dissemination.
- Developed an open, self-paced Canvas course by curating recordings, structuring learning modules, and aligning resources to support asynchronous, scalable faculty learning nationwide.
- Supported certificate-of-completion workflows for both synchronous and asynchronous participation, contributing to participant recognition and institutional visibility as a national convener of AI in higher education.

Project: The Advanced Infrastructure, Energy, and Agriculture (AIREA) Data Explorer (URL: [AIREA Data Explorer](#))

University of Tennessee | Funded by JP Morgan Chase

PI: Dr. Cameron Sublett | [Project Site](#)

R Shiny Repository: <https://github.com/data-edu/airea-data-explorer-shiny>

Data Prep Repository: <https://github.com/data-edu/airea-data-explorer-prep>

- Designed and deployed the *AIREA Data Explorer*, a web-based visualization platform showcasing how U.S. community colleges prepare students for infrastructure, energy, agriculture, and related high-opportunity occupations in infrastructure, energy, and agriculture.
- Built the platform using R Shiny and Mapbox GL JS, integrating national datasets to visualize changes in job opportunities and credential completions since 2010.
- Collaborated with researchers at the Community College Research Center (Columbia University) to ensure research rigor, data accessibility, and user-centered interface design.
- The dashboard allows policymakers, educators, and researchers to analyze alignment between community college programs and labor market needs, promoting data-driven educational planning.
- Serves as a model for open, reproducible data communication in applied educational research and workforce analytics.

Institute of Education Sciences (IES) Funded Project — Quantifying the Robustness of Causal Inferences

Award No. R305D220022 | Michigan State University (Collaborative)

PI: Dr. Kenneth Frank | [IES Project Info](#)

- Contributed to the development and validation of the “[Konfound](#)” R package, an open-source statistical tool for causal inference robustness analysis.
- Collaborated with cross-institutional teams to test algorithms, ensure code reproducibility, and co-author peer-reviewed publications in *Journal of Open Source Software (JOSS)*.
- Enhanced model evaluation procedures relevant to AI-based educational analytics

and scalable statistical workflows.

Project: Individualized Teaching and Educational Governance Supported by Artificial Intelligence

iFLYTEK Education Technology Institute | Bengbu, China | Grant No. IED2020-Z003

PI: Dr. Bang Qi Liu

- Conducted field studies on AI-supported educational governance and school-level adoption of smart classroom technologies.
- Designed and implemented qualitative interview protocols for teachers and principals, then coded and analyzed data to assess the impact of AI-driven teaching models.
- Authored regional governance reports that informed provincial education policy and smart classroom development strategies.

Project: Anhui Philosophy and Social Science Planning Project — Knowledge Supply and Deep Learning in the “Internet+” Era.

Anhui Normal University | Grant No. AHSKY2019D036

PI: Dr. Hehai Liu

- Developed and validated a survey instrument grounded in the ECM-ISC model to measure students’ continued engagement with online learning environments.
- Applied multilevel regression and confirmatory factor analysis using R and SPSS to assess behavioral intention and technology adoption factors.
- Published as first author in peer-reviewed journals, contributing to research on digital learning adoption and instructional technology design.

WORK EXPERIENCE

Digital Learning Unit, University of Tennessee, Knoxville. USA.

Teaching Assistant & Research Assistant | *Aug 2023 – Present*

- Conducted research on AI-enhanced online teaching practices, synthesizing empirical evidence to inform design recommendations for higher-education contexts.
- Designed and administered survey and interview protocols; led qualitative coding and mixed-method analysis on faculty perceptions and adoption of AI tools.
- Facilitated faculty “Brown Bag” sessions and analyzed feedback to improve program quality and resource alignment.
- Collaborated with a research team to prototype AI-informed instructional frameworks (e.g., teacher support workflows and reusable design patterns) and document implementation guidance.

National Cognitive Intelligence Key Laboratory, Institute of Educational Technology, iFLYTEK. China.

Researcher | *Mar 2021 – Jun 2023*

- Conducted research in Smart Education Demonstration Zones, evaluating AI-enabled learning platforms and K–12 smart classroom ecosystems across regional pilots.
- Led mixed-method evaluation studies to examine impacts on teaching practices and student engagement (learning analytics + surveys/interviews/classroom observations).
- Developed and maintained an education policy and initiative database on intelligent education, supporting evidence-based reporting and consultation for local education authorities.
- Contributed to the *China Annual Blue Book of Artificial Intelligence Education Development* by synthesizing national findings and translating implementation insights into actionable recommendations.
- Coordinated research-practice communication through the Annual Intelligent Education Development Forum, engaging researchers, educators, and product teams.

Shanghai Squirrel AI Education Technology Co., Ltd. China
Instructor, Adaptive Learning Specialist | *Dec 2017 – Aug 2019*

- Implemented adaptive learning workflows for K–12 English learners using diagnostic and recommendation systems to personalize practice pathways.
- Supported product validation by collecting classroom-side observations and learner performance signals, and collaborating with teams to iterate on adaptive features.
- Delivered teacher training on technology integration, digital pedagogy, and information literacy, translating system capabilities into classroom-ready practices.

CONFERENCE PRESENTATIONS

- **Wang, W** (2026). Localized Large Language Models for Educational Research: Two Privacy-Preserving Case Studies in Vision and Text Analysis. Chicago, AECT 2026.
- **Wang, W** (2026). Bridging Workforce Data and Decision-Making with a RAG-Enabled Community College Data Explorer. Stanford, Stanford EDS 2026.
- **Wang, W** (2026). Developing Prompt Engineering Competence Framework: Comparing China, EU, and U.S. Educators' Digital Literacy Standards. Los Angeles, AERA 2026.
- **Wang, W** (2026). Educational Video Transcript Analysis with LLMs: Improving Entity Recognition and Qualitative Insights. Los Angeles, AERA 2026.
- **Wang, W** (2025). Bridging the AI Gap: Collaborative Strategies for Effective Integration in Higher Education. Orlando, OLC.2025.
- **Wang, W** (2025). Aligning Degree Production and In-Demand Middle Skill Occupations: National and Regional Insights. Seattle, APPAM.2025.
- **Wang, W** (2025). Faculty Voices: Sentiment and Thematic Analysis of GenAI Integration in Higher Education Course Design. Las Vegas, AECT.2025.
- Hongjie Ping, **Wang, W** (2025). Generative AI Dependency in Higher Education: Investigating Continuance Intention, Cognitive Response, and Creativity. The 7th International Conference on Computer Science and Technologies in Education (CSTE). Wuhan, China. CSTE 2025.
- **Wang, W** (2025). Exploring Dimensions of AI Integration and Designing an Effective GenAI-Based Teaching Model. Denver, AERA.2025.
- **Wang, W.** (2024). Navigating Sentimental Landscapes: An In-depth Analysis of ChatGPT Implementation in Universities through Policy Examination. Philadelphia, AERA.2024.
- **Wang, W.** (2024). Implementing Generative AI in Practice: Designing Assessments and Learning Activities in Faculty Development Program. Kansas City, AECT .2024.

SERVICE

Peer Review, Journal of Research on Technology in Education, 2024.
Peer Review, Journal of Research on Technology in Education, 2025.
Peer Review, Journal of Statistics and Data Science Education, 2025.
Peer Review, AECT 2025.
President of CSSA, Student Union, University of Tennessee.

PROFESSIONAL SKILLS & RECOGNITIONS

Technical & Analytical Skills

- Programming Languages: Python, R
- AI/ML Frameworks: PyTorch, Hugging Face, OpenAI API, Scikit-learn

- Statistical & Modeling Tools: RStudio, SPSS, Mplus, Amos
- Data Science: Educational data mining, LLM evaluation, causal inference, NLP applications
- Data Visualization: R Shiny, Mapbox GL JS, Tableau, ggplot2
- Version Control & Cloud: Git/GitHub, AWS, Google Cloud
- Web & App Development: HTML/CSS, JavaScript, mobile and web-based educational platform design

Research & Design Skills

- Instructional Design: Applying design-based research for AI-supported teaching and learning environments
- Empirical Research: Mixed-methods data collection, large-scale survey design, factor analysis, and regression modeling
- AI for Education: Integration of large language models (LLMs) and generative AI tools in classroom and faculty development contexts
- Evaluation & Assessment: Quantitative and qualitative evaluation of educational technologies and instructional effectiveness
- Publication & Dissemination: Experience co-authoring peer-reviewed articles, technical reports, and public data dashboards

Certificates

- National English Teacher Qualification Certificate (Junior High, China)
- National Mathematics Teacher Qualification Certificate (Junior High, China)
- National IT Teacher Qualification Certificate (Senior High, China)

Awards & Honors

- Julius D. McElroy Scholarship, University of Tennessee, 2023
- GrowSTEM Project Fund Scholarship, University of Tennessee, 2023
- Outstanding Master's Thesis Award, Anhui Normal University, 2022